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Contact Multi-point Photoelectric Liquid Level Sensor SKU: CQRSENYW003

Ambient Light Sensor

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Description

Contact Multi-point Liquid Level Sensor, This is a photoelectric water liquid level sensor that is operates using optical principles. Open collector output mode, suitable for connecting various circuits and product applications.

The sensor has no mechanical parts, requires no additional adjustment, and has high sensitivity, low power consumption, corrosion resistance, high pressure resistance, high temperature resistance and chemical stability.

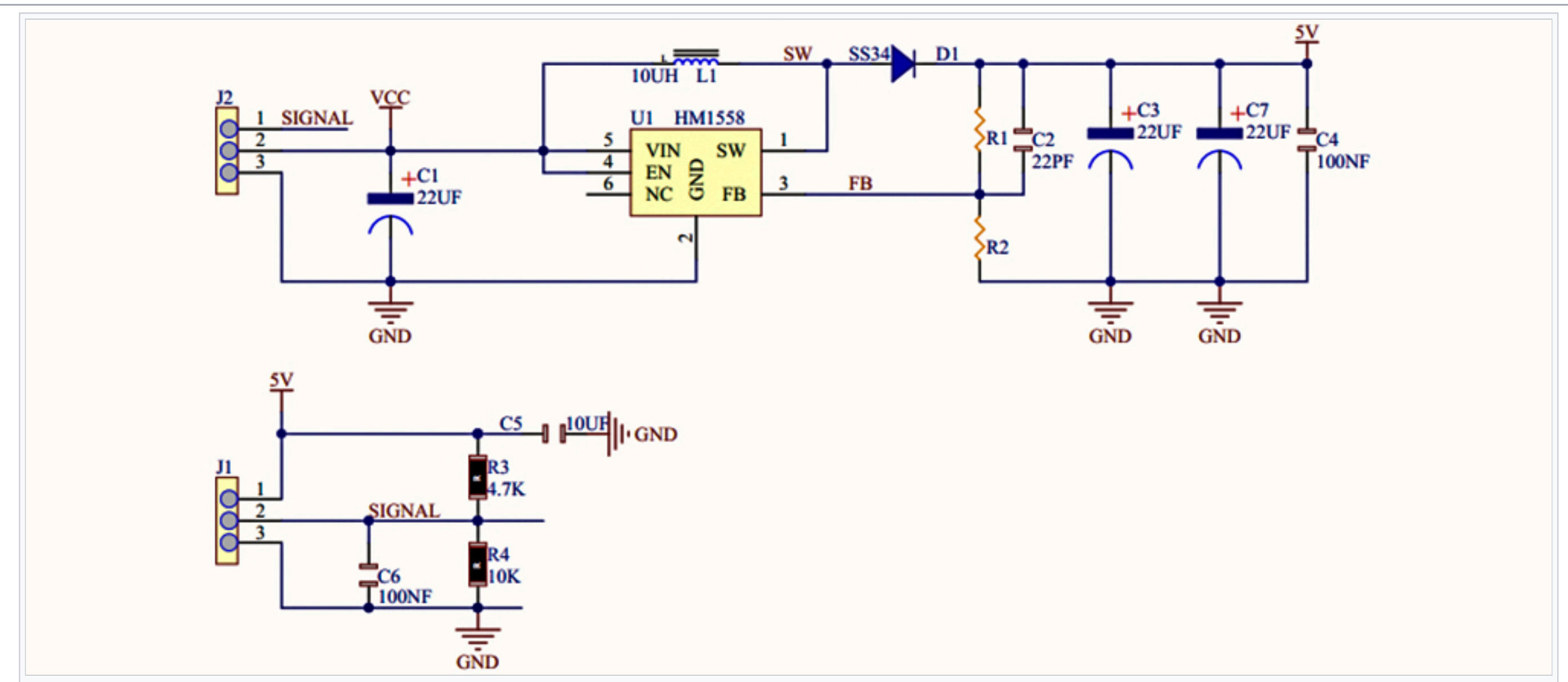
This sensor probe is small in size and has a structure that can be placed up, down, laterally, and diagonally in multiple orientations to detect solution spillage, dryness and horizontal level. Can be used as a reminder and alarm function.

The multi-point liquid level sensor can detect 4 liquid levels with a detection accuracy of up to ±1 mm, high reliability and ultra-low standby power consumption.

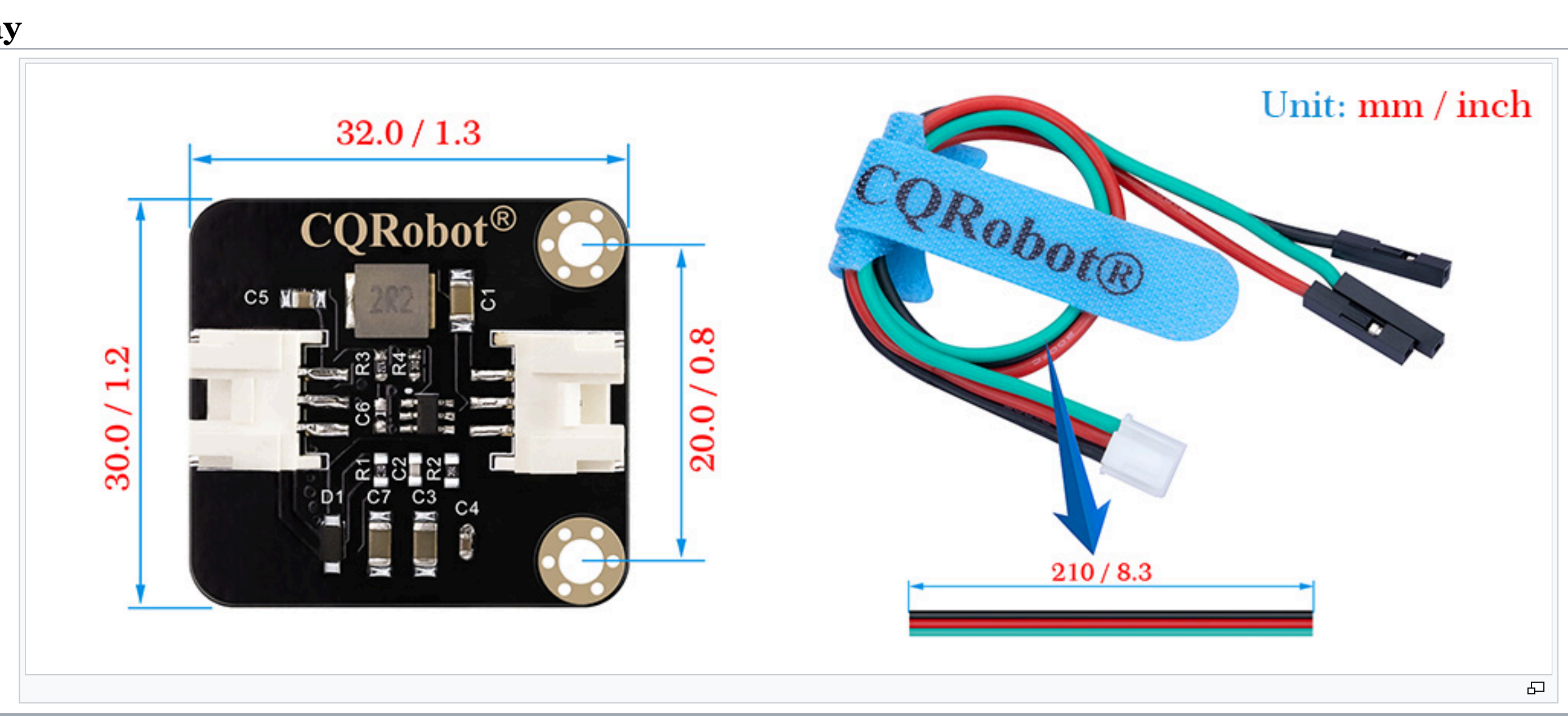
Compatible with Arduino motherboard and Raspberry Pi motherboard. for Automatic Irrigation Systems, Aquariums, Plants, in The Garden, in Agriculture etc.

Note: Avoid placing the sensor near bright lights or in direct sunlight as these can cause interference.

Product Principle Diagram



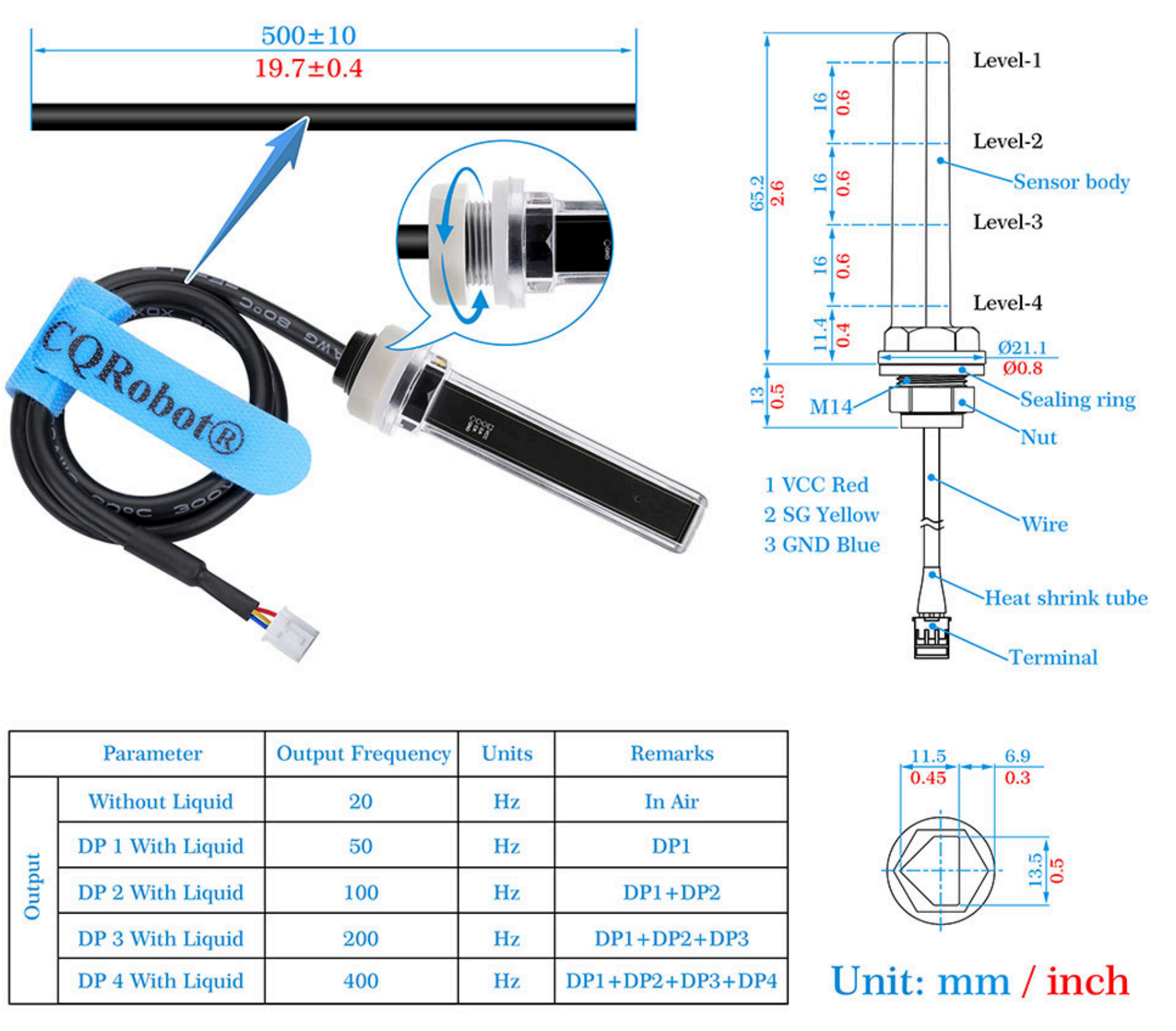
Size Display



Specifications

Contact Multi-point Photoelectric Liquid Level Sensor

- Model: CQRSENYW003;
- Working Voltage: 3.3V / 5V;
- Working Current: 3.3V / minimum 80mA;
- Working Current: 5V / minimum 55mA;
- Output Frequency Without Liquid: 20Hz;
- Output Frequency Detection Point 1 With Liquid: 50Hz;
- Output Frequency Detection Point 2 With Liquid: 100Hz;
- Output Frequency Detection Point 3 With Liquid: 200Hz;
- Output Frequency Detection Point 4 With Liquid: 400Hz;
- Working Temperature: -10 Degree Celsius to +60 Degree Celsius;
- Dimensions: 32 mm * 30 mm;
- Mounting Hole Size: 3.0 mm;



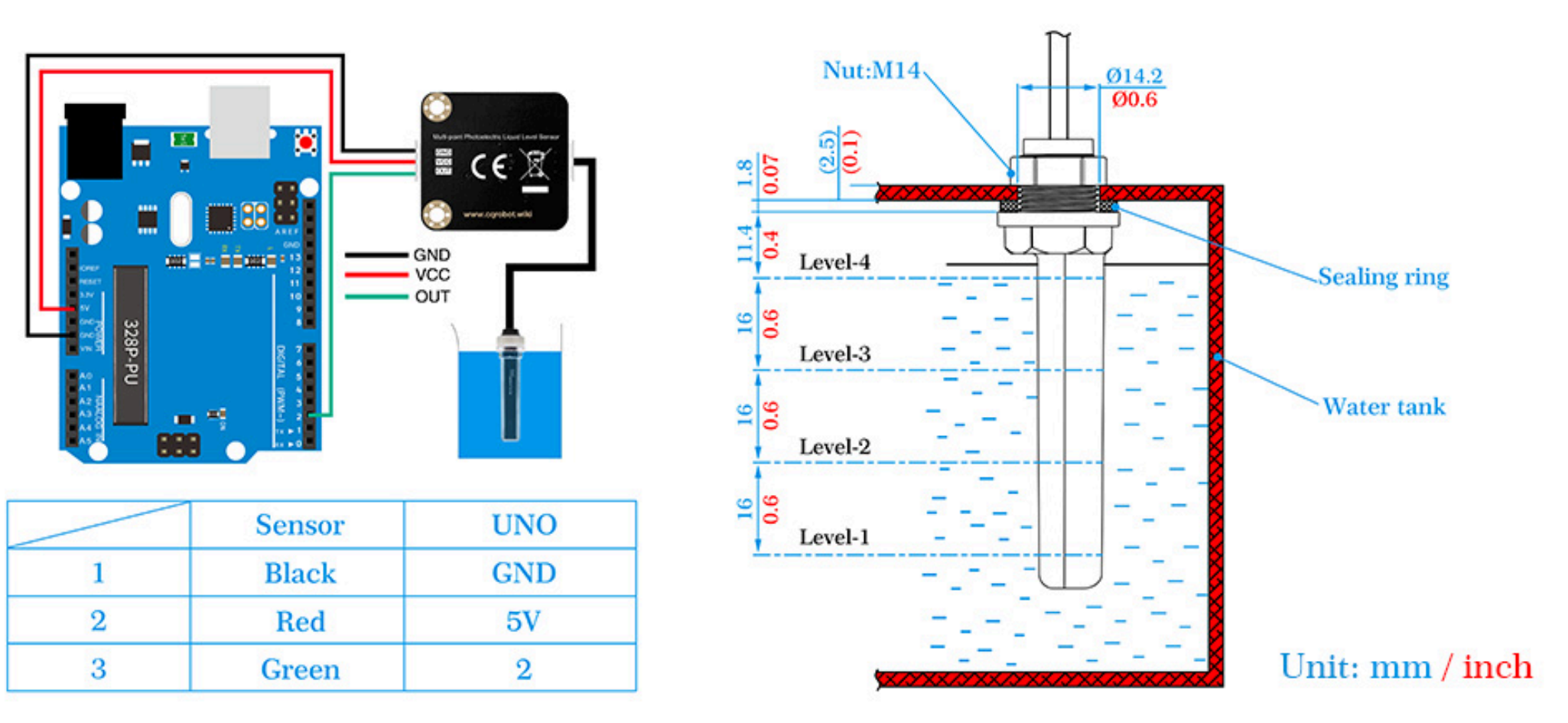
Parameter		Output Frequency	Units	Remarks
Output	Without Liquid	20	Hz	In Air
	DP 1 With Liquid	50	Hz	DP1
	DP 2 With Liquid	100	Hz	DP1+DP2
	DP 3 With Liquid	200	Hz	DP1+DP2+DP3
	DP 4 With Liquid	400	Hz	DP1+DP2+DP3+DP4

Cable Specifications

- Cable Specifications: 22AWG;
- Material: Silicone;
- Withstand Voltage: Less Than 50V;
- Withstand Current: Less Than 1000MA;
- Length: 21cm;
- Line Sequence: Black-Negative Power Supply, Red-Positive Power Supply, Green-Output Frequency.

Arduino Example and Test Code

1. Arduino Connection Method



	Sensor	UNO
1	Black	GND
2	Red	5V
3	Green	2

2. Sample Code

2.1 Software download link: https://www.arduino.cc/en/software

2.2 Download and Run the Test Examples

Media: CQRSENYW003-Arduino.rar

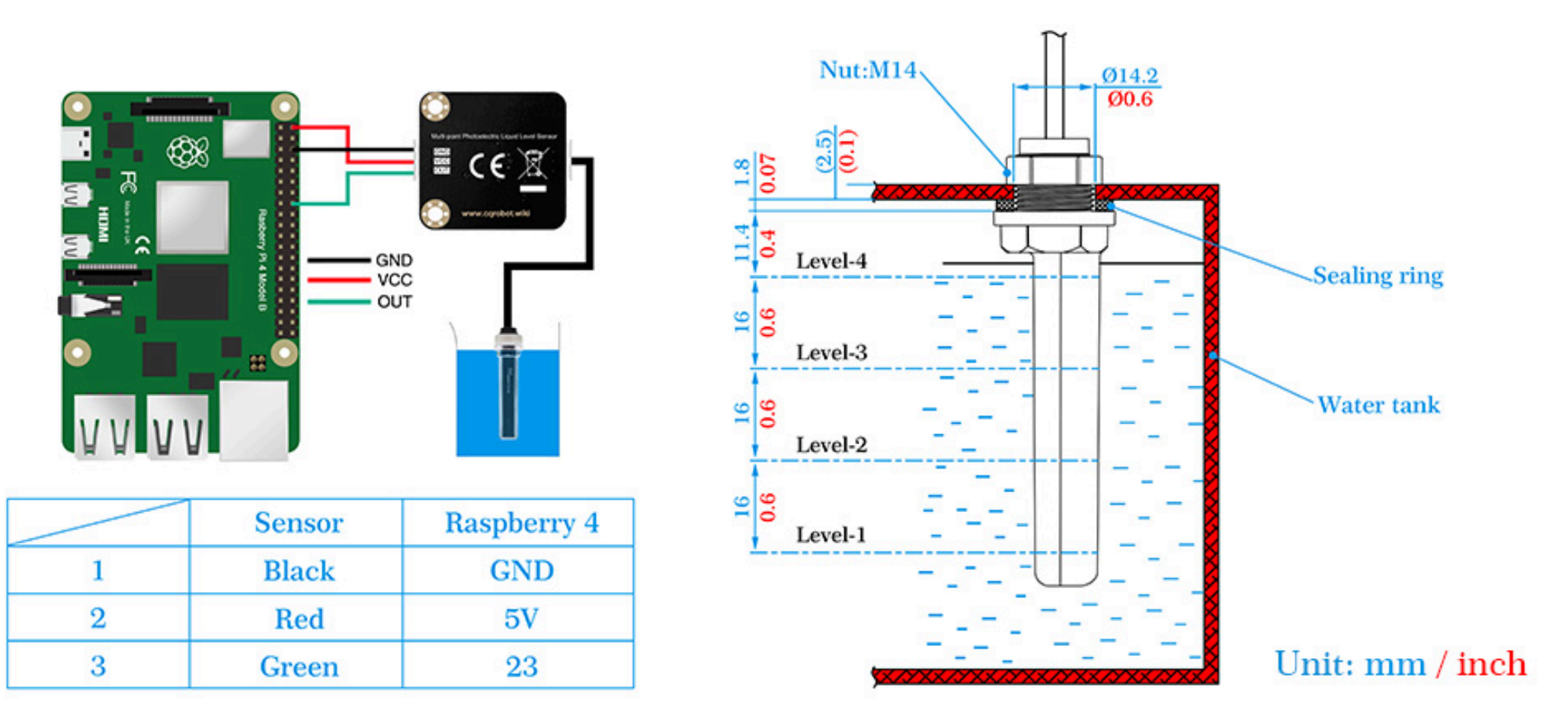
```
long time1, time2, time3;
float Frequency;

void setup()
{
  Serial.begin(9600);                   //Initialize the serial port
  pinMode(2, INPUT_PULLUP);             //Set D2 to pull up
  attachInterrupt(0, Blink, FALLING);  //Set interrupt 0 (UNO digital D2)
}

void loop()
{
  //Print frequency value through serial port
  Serial.print("Frequency=");
  Serial.println(Frequency + String(" HZ"));
  delay(100);
}

//Interrupt function
void Blink()
{
  time1 = time2;
  time2 = micros();
  time3 = time2 - time1;
  Frequency = 1000000.0 / time3;
}
```

3. Example

Upload the program to the Arduino UNO board and connect the wires correctly. Open the serial port and set the baud rate to 9600. When the sensor is not in contact with liquid, the serial port prints Frequency = 20 HZ; when the water level is at Level-4 (1) in the figure below, the serial port prints Frequency = 50 HZ; when the water level is at Level-3 (2) in the figure below, the serial port prints Frequency = 100 HZ; when the water level is at Level-2 (3) in the figure below, the serial port prints Frequency = 200 HZ; when the water level is at Level-1 (4) in the figure below, the serial port prints Frequency = 400 HZ.

Raspberry Pi Example and Test Code

1. Raspberry Pi Connection Method



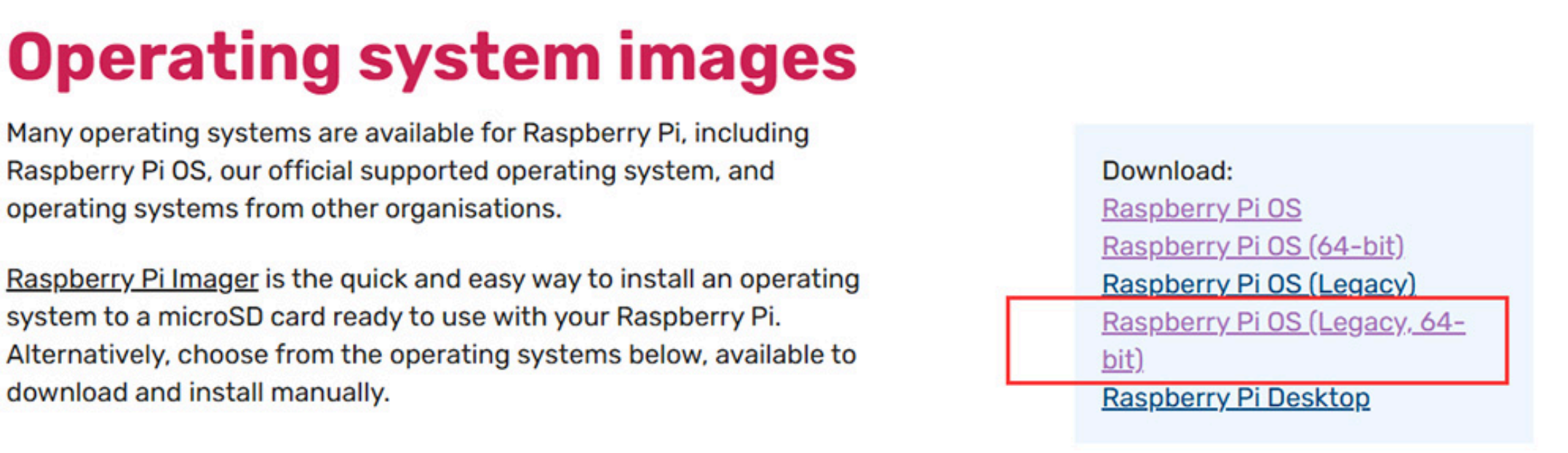
	Sensor	Raspberry 4
1	Black	GND
2	Red	5V
3	Green	23

2. Sample Code

2.1 Software download link:

First, go to the Raspberry Pi official website to download the system: You must select the download box in the figure below, otherwise the sensor will not work properly.

https://www.raspberrypi.com/software/operating-systems/#raspberry-pi-os-legacy-64-bit

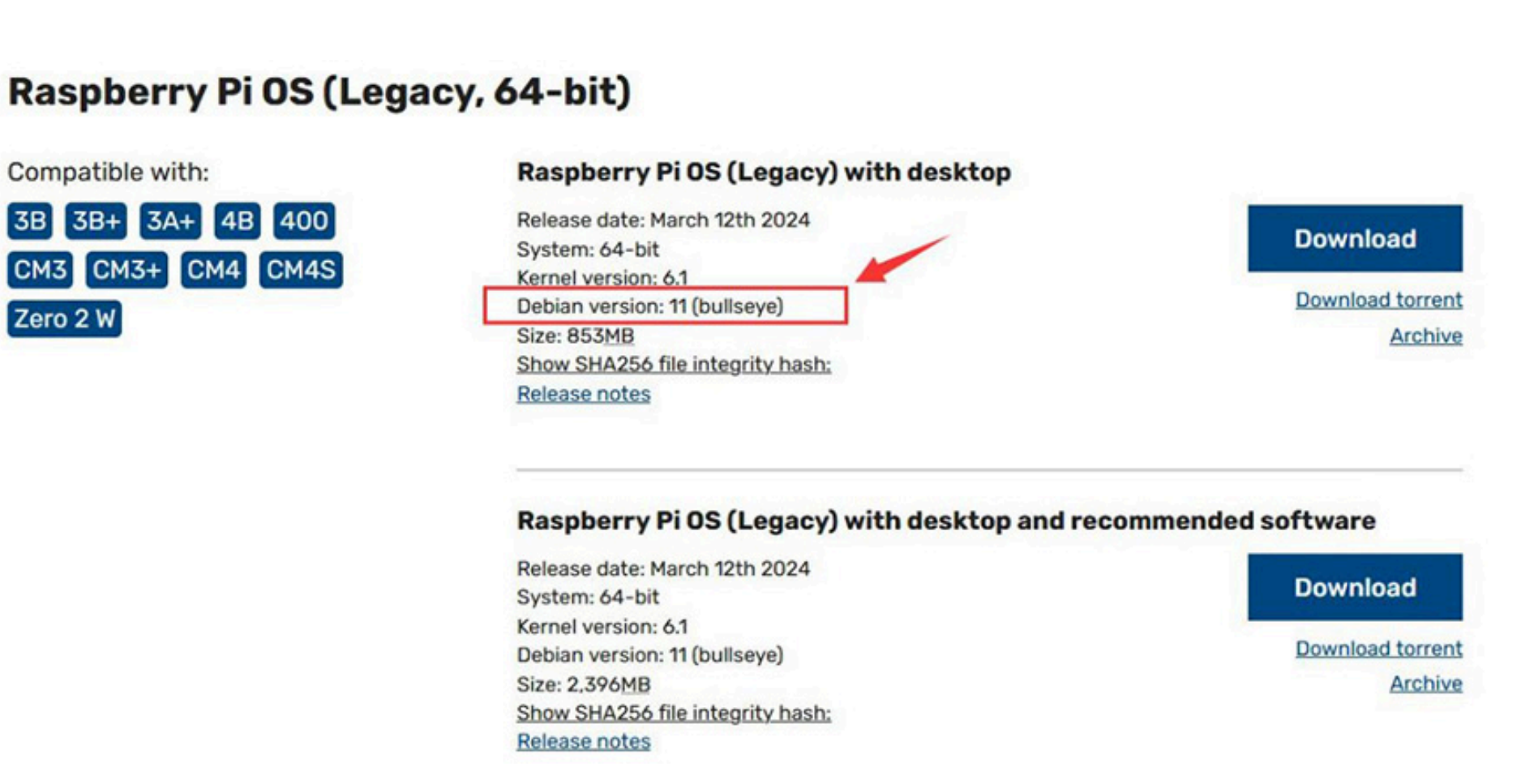


Operating system images

Many operating systems are available for Raspberry Pi, including Raspberry Pi OS, our official supported operating system, and operating systems from other organisations.

Raspberry Pi Imager is the quick and easy way to install an operating system to a microSD card ready to use with your Raspberry Pi. Alternatively, choose from the operating systems below, available to download and install manually.

Download:
Raspberry Pi OS
Raspberry Pi OS (64-bit)
Raspberry Pi OS (Legacy)
Raspberry Pi OS (Legacy, 64-bit)
Raspberry Pi Desktop



Raspberry Pi OS (Legacy, 64-bit)

Compatible with:
3B 3B+ 3A+ 4B 400 CM3 CM3+ CM4 CM4S Zero 2 W

Raspberry Pi OS (Legacy) with desktop
Release date: March 12th 2024
System: 64-bit
Kernel version: 6.1
Debian version: 11 (bullseye)
Size: 853MB
Show SHA256 file integrity hash:
Release notes
Download
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Raspberry Pi OS (Legacy) with desktop and recommended software
Release date: March 12th 2024
System: 64-bit
Kernel version: 6.1
Debian version: 11 (bullseye)
Size: 2,396MB
Show SHA256 file integrity hash:
Release notes
Download
Download torrent
Archive

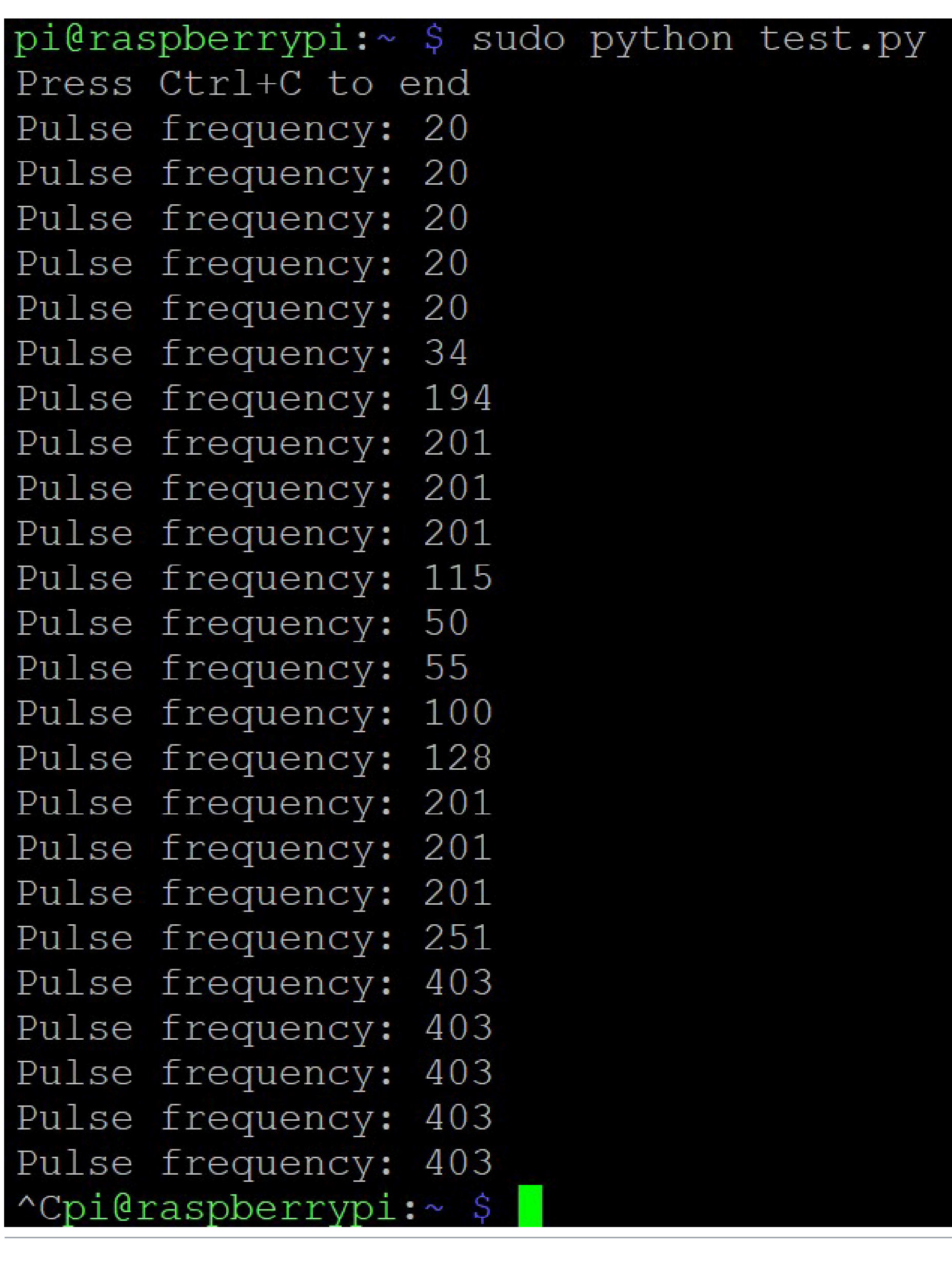
2.2. Download and Run the Test Examples

Media: CQRSENYW003-Raspberry Pi.rar

```
import RPi.GPIO as GPIO
from time import time, sleep

GPIO.setmode(GPIO.BCM)
GPIO.setwarnings(False)

GPIO_IN = 23
GPIO.setup(GPIO_IN, GPIO.IN, pull_up_down=GPIO.PUD_UP)

pulse_count = 0
last_time = 0
total_time = 0
last_time = time()

def handle_counter(channel):
    global pulse_count, last_time, total_time
    pulse_count += 1
    total_time = total_time + (time() - last_time)
    last_time = time()

GPIO.add_event_detect(GPIO_IN, GPIO.RISING, callback=handle_counter)

try:
    print("Press Ctrl+C to end")
    while True:
        sleep(1)
        f = pulse_count/total_time
        print("Pulse frequency: %d" %(f))
        pulse_count = 0
        total_time = 0
except KeyboardInterrupt:
    pass

GPIO.cleanup()
```

3. Example

Save the program to the Raspberry 4 development board and connect the wires correctly. When you run the program by typing sudo python test.py, you can see the following figure. When the sensor is not in contact with the liquid, it prints Pulse Frequency: 20; when the water level is at Level-4 (1) in the figure below, it prints Pulse Frequency: 50; when the water level is at Level-3 (2) in the figure below, it prints Pulse Frequency: 100; when the water level is at Level-2 (3) in the figure below, it prints Pulse Frequency: 200; when the water level is at Level-1 (4) in the figure below, it prints Pulse Frequency: 400.



```
pi@raspberrypi:~ $ sudo python test.py
Press Ctrl+C to end
Pulse frequency: 20
Pulse frequency: 20
Pulse frequency: 20
Pulse frequency: 20
Pulse frequency: 20
Pulse frequency: 34
Pulse frequency: 194
Pulse frequency: 201
Pulse frequency: 201
Pulse frequency: 201
Pulse frequency: 115
Pulse frequency: 50
Pulse frequency: 55
Pulse frequency: 100
Pulse frequency: 128
Pulse frequency: 201
Pulse frequency: 201
Pulse frequency: 201
Pulse frequency: 251
Pulse frequency: 403
Pulse frequency: 403
Pulse frequency: 403
Pulse frequency: 403
Pulse frequency: 403
^Cpi@raspberrypi:~ $
```

Common Problem

The sensor does not work

- 1. Make sure the sensor wire is connected to the development board correctly.
- 2. Then check whether the program of the development board is burned correctly.

This page was last edited on 9 August 2024, at 11:05.

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